

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q22-2008

Sorcey Rd 100 kV  
December 25, 2023

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2008. The Interconnection Customer proposes a 46.4 MW battery energy storage facility interconnecting at the Sorcey Rd 100 kV substation in the Ohio area. The request was received on February 16, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 506 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,940,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: AES – KNOX 230.0 kV line  
Contingency: ATASCOCITA – NORTH CAMBRIDGE 230.0 kV line  
Loading: 108.8% (Rating: 492 MVA)  
DFAX: 6.0%  
Pre-project loading: 108.2%
2. Facility: DOVER – WEST NEW PHILADELPHIA 100.0 kV  
Contingency: BROWNTOWN – LOCKHEED 2 TAP 100.0 kV  
Loading: 106.9% (Rating: 100 MVA)  
DFAX: 26.7%  
Pre-project loading: 94.6%
3. Facility: W BELLAIRE SUBSTATION 138-1 kV Autotransformer  
Contingency: W BELLAIRE SUBSTATION – SPRINGHILL 230.0 kV  
Loading: 104.2% (Rating: 235 MVA)  
DFAX: 7.9%  
Pre-project loading: 102.7%

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

### **4. Required Network Upgrades and Cost Estimates**

| Facility                                       | Required Upgrade                           | Cost Estimate | Allocation to Project |
|------------------------------------------------|--------------------------------------------|---------------|-----------------------|
| AES – KNOX 230.0 kV                            | Reconductor line                           | \$6,947,000   | 11% (\$741,000)       |
| DOVER – WEST NEW PHILADELPHIA 100.0 kV         | Reconductor line                           | \$1,525,000   | 61% (\$931,000)       |
| W BELLAIRE SUBSTATION 138-1 kV Autotransformer | Replace transformer with higher-rated unit | \$3,268,000   | 100% (\$3,268,000)    |
|                                                | Total allocated to project                 |               | \$4,940,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.